Part IX — Modern Physics and the Universe · Chapter 48

Atoms and Spectra

How light reveals the hidden energy structure of matter

> *Hold a piece of glass prism to sunlight and the white beam fans into a rainbow of colour. Heat a piece of sodium in a flame and it glows with a brilliant yellow that is unmistakably its own. Aim a spectrometer at a distant star and read off a barcode of dark lines that tells you the star is made of hydrogen, helium, and iron — without leaving Earth. These are not decorative curiosities. They are windows into the interior of atoms. Every line in every spectrum is a message written by electrons shifting between precisely defined energy levels inside a specific atom. Understanding atoms and spectra means understanding why matter is stable, why elements are distinct, and why energy is not continuous but quantised. This chapter builds the conceptual foundation for all of modern physics.*

48.1 Why Atoms Matter in Modern Physics

Classical physics — Newton's mechanics, Maxwell's electromagnetism, thermodynamics — was triumphantly successful by the end of the nineteenth century. It described the motion of planets, the behaviour of engines, the propagation of light, and the flow of electricity with extraordinary precision. But it failed completely when applied to atoms.

The problem was stability. Classical electromagnetism predicted that an electron orbiting a nucleus would continuously radiate energy (an accelerating charge emits radiation), spiral inward, and collapse into the nucleus in about 10 nanoseconds. Matter as we know it — stable, with definite chemical properties — would be impossible. Yet atoms clearly are stable. Chemistry works. Solid objects exist.

Beyond stability, classical physics could not explain the discrete colours of light emitted by hot gases. A hot solid emits a continuous rainbow of light. A hot gas emits only specific, sharp wavelengths — a set of isolated colours unique to each element. Classical physics had no mechanism to produce this. The explanation required a completely new idea: that electrons inside atoms can only occupy specific allowed energy states, and that they absorb and emit light only in discrete packets corresponding to jumps between those states.

The study of atoms and spectra is therefore not merely historical background. It is the entry point into quantum physics — the most accurate and far-reaching physical theory ever developed.

48.2 Early Atomic Models

The idea that matter is made of indivisible particles — atoms — is ancient, proposed by Democritus in ancient Greece around 400 BCE. But the scientific theory of atoms began with John Dalton's work in 1803–1808. Dalton proposed that each element consists of identical atoms, that atoms of different elements differ in mass, and that chemical reactions rearrange atoms without destroying or creating them. This explained the law of conservation of mass and the law of definite proportions.

> **Historical Note:** John Dalton (1766–1844) was a Quaker schoolteacher and meteorologist whose work on atmospheric gases led him to atomic theory. He produced the first table of relative atomic masses. His atoms were solid, indivisible spheres — a "billiard ball" model with no internal structure.

Dalton's model was revised fundamentally when J. J. Thomson discovered the electron in 1897. By measuring the charge-to-mass ratio of cathode rays — beams of particles emitted from metals in high-vacuum tubes — Thomson showed that atoms contain negatively charged particles (electrons) much lighter than any atom. Since atoms are electrically neutral, they must also contain positive charge.

Thomson proposed the "plum pudding" model (1904): the atom is a diffuse sphere of uniformly distributed positive charge (the "pudding") with electrons embedded throughout (the "plums"). This model was reasonable but wrong. It was disproved within a decade by Rutherford's scattering experiment.

Model	Proposed by	Year	Key Idea	Fatal Flaw
Solid sphere	Dalton	1803	Indivisible solid atoms	No internal structure cannot explain electrons o spectra
Plum pudding	Thomson	1904	Electrons in diffuse positive sphere	Disproved by Rutherfor scattering (1909)
Nuclear model	Rutherford	1911	Dense positive nucleus; electrons orbit outside	Classical prediction electron spirals int nucleus in ~10 ns
Bohr model	Bohr	1913	Electrons in fixed allowed orbits; quantised angular momentum	Works only for hydrogen no explanation for why orbits are stable
Quantum model	Schrödinger / Heisenberg	1925–1926	Electrons described by probability wave functions	Correct; basis of moderr chemistry and physics

Table 48.1 *Historical development of atomic models, each superseded by new evidence.*

48.3 The Nuclear Model of the Atom

In 1909, Hans Geiger and Ernest Marsden, working under Ernest Rutherford at the University of Manchester, fired a beam of alpha particles (helium nuclei, charge +2e) at an extremely thin gold foil. According to the Thomson model, the positive charge was spread uniformly through the atom, so alpha particles should pass through with only minor deflections — like bullets through a cloud of fog.

The results were shocking. Most alpha particles did pass straight through, but a small fraction — about 1 in 8000 — were deflected by large angles, some bouncing almost directly backward. Rutherford famously said it was "as surprising as if you fired artillery shells at tissue paper and they came back and hit you."

Rutherford's interpretation (1911): the positive charge of the atom is concentrated in an extraordinarily small, dense region at the centre — the nucleus. The nucleus is roughly 10^{-15} m in diameter, while the atom itself is roughly 10^{-10} m. The nucleus is 100,000 times smaller than the atom — like a marble at the centre of a football stadium. The electrons orbit the nucleus in the surrounding empty space.

> **Key Point:** The nuclear atom is almost entirely empty space. If the nucleus were scaled up to the size of a grain of sand (1 mm), the nearest electron would be 100 m away. This is why alpha particles mostly pass through gold foil undeflected: they travel through the vast empty space between nuclei. The rare large deflections occur when an alpha particle passes close to a nucleus.

Rutherford's model correctly explained the scattering experiment but failed to explain why orbiting electrons do not radiate and collapse. It also could not explain spectral lines. These problems were solved — partially — by Bohr.

48.4 Electrons and Atomic Structure

In Rutherford's nuclear model, the electrons are outside the nucleus. For a hydrogen atom, there is one proton (charge $+e$) in the nucleus and one electron (charge $-e$) orbiting it. For helium, there are two protons and two electrons. For carbon, six protons and six electrons.

The atomic number Z is the number of protons in the nucleus and equals the number of electrons in a neutral atom. It determines the element: $Z = 1$ is hydrogen, $Z = 6$ is carbon, $Z = 79$ is gold. The chemical properties of an element depend almost entirely on the arrangement of its electrons, because chemical reactions involve interactions between the electron clouds of atoms.

The nucleus also contains neutrons (electrically neutral particles discovered by James Chadwick in 1932). Neutrons contribute to the nuclear mass but not to the atomic number or chemical properties. The mass number A is the total number of protons and neutrons: $A = Z + N$ where N is the number of neutrons.

The electron has a negative charge of magnitude $e = 1.602 \times 10^{-19}$ C and a rest mass of $m_e = 9.109 \times 10^{-31}$ kg — about 1/1836 the mass of a proton. Because electrons are so much lighter than protons and neutrons, essentially all the mass of an atom is in the nucleus.

48.5 Energy Levels in Atoms

The key insight of modern atomic physics is that electrons in atoms do not occupy just any orbit or energy: they are restricted to a set of discrete, allowed energy states called energy levels. This is the quantum nature of the atom.

In classical mechanics, a planet orbiting the sun can orbit at any distance and with any energy — there is a continuous range of possible orbits. In an atom, only specific energies are permitted. These allowed states are indexed by quantum numbers ($n = 1, 2, 3, \ldots$), where $n = 1$ is the ground state (lowest energy, most stable) and higher values of n are excited states (higher energy).

The energy levels of hydrogen (the simplest atom, with one proton and one electron) are given by the Bohr formula:

$$E_n = -13.6 / n^2 \quad \text{eV}$$

Hydrogen energy levels. E_n = energy of level n in electron-volts (eV); $n = 1, 2, 3, \ldots$ is the principal quantum number. The ground state ($n = 1$) has energy −13.6 eV. The negative sign means the electron is bound to the nucleus (energy must be added to free it).

The ground state energy of −13.6 eV is the ionisation energy of hydrogen: it takes 13.6 eV of energy to remove the electron from a hydrogen atom in its ground state. As n increases, the energy levels become more closely spaced and approach 0 eV (the free electron limit) from below.

Level (n)	Energy (eV)	Energy (J)	Name
1	−13.60	-2.18×10^{-18}	Ground state
2	−3.40	-5.45×10^{-19}	First excited state
3	−1.51	-2.42×10^{-19}	Second excited state
4	−0.85	-1.36×10^{-19}	Third excited state
5	−0.54	-8.72×10^{-20}	Fourth excited state
∞	0	0	Ionisation limit (free electron)

Table 48.2 *Energy levels of the hydrogen atom. Energies are negative because the electron is bound.*

48.6 Quantised Energy

The word “quantised” means that a physical quantity is restricted to specific discrete values rather than varying continuously. The energy of an electron in an atom is quantised: it can be −13.6 eV, −3.4 eV, −1.51 eV, or other allowed values, but never −6.0 eV or −2.0 eV or any other value between allowed levels.

This is completely unlike classical physics. A ball rolling down a ramp can have any energy from zero to its initial potential energy — the energy varies continuously as it rolls. A planet can orbit at any distance and energy. But an electron in a hydrogen atom is forced to jump discontinuously between allowed levels. It cannot gradually slide from one level to another.

Quantisation arises from the wave nature of electrons (discussed in Chapter 49). An electron in an atom must form a standing wave that fits around the nucleus — only certain wavelengths fit, corresponding to the discrete allowed energy levels. This is analogous to the allowed resonant frequencies of a guitar string, which can vibrate at its fundamental frequency or its harmonics but not at arbitrary frequencies in between.

> **The Rule:** An electron in an atom can only exist at an allowed energy level. It cannot have an energy between levels. To move to a higher level, it must absorb exactly the right amount of energy. To drop to a lower level, it must release exactly the right amount of energy, typically as a photon of light.

48.7 Emission Spectra

When atoms are excited — by heating a gas, passing an electrical current through it, or illuminating it — their electrons absorb energy and jump to higher energy levels. These excited states are unstable. After a very short time (typically 10^{-8} to 10^{-6} seconds), the electron drops back to a lower energy level. The energy lost in this transition is released as a photon of light.

The energy of the emitted photon exactly equals the difference in energy between the two levels:

$$E\_photon = E\_upper - E\_lower = hf = hc / \lambda$$

Photon energy for an atomic transition. h = Planck's constant = 6.626×10^{-34} J·s; f = frequency (Hz); c = 3.00×10^{8} m/s; λ = wavelength (m). Note 1 eV = 1.602×10^{-19} J.

Because only specific energy differences are possible (determined by the allowed energy levels), only specific photon energies are emitted. Since each photon energy corresponds to a specific frequency, and each frequency corresponds to a specific colour (wavelength), the atom emits only specific colours of light. This produces the emission spectrum — a set of bright spectral lines against a dark background.

Different elements have different energy levels, so different elements emit different sets of spectral lines. The emission spectrum of each element is unique — a fingerprint that identifies that element unambiguously. Hydrogen emits a distinctive pattern including a red line (656 nm), a blue-green line (486 nm), and two violet lines. Sodium emits a pair of intense yellow lines near 589 nm. Neon emits many red and orange lines (which is why neon signs glow red-orange).

Example — Photon energy from a hydrogen transition An electron in hydrogen drops from n = 3 to n = 2. $\Delta E = E\_3 - E\_2 = (-1.51) - (-3.40) = 1.89$ eV. Convert: $1.89 \times 1.602 \times 10^{-19} = 3.03 \times 10^{-19}$ J. Wavelength: $\lambda = hc/E = (6.626 \times 10^{-34} \times 3.00 \times 10^{8}) / (3.03 \times 10^{-19}) = 656$ nm. This is the red H-alpha line, the most prominent line in hydrogen's visible spectrum.

48.8 Absorption Spectra

The reverse process also occurs. If white light (containing all wavelengths) is passed through a cool gas, the atoms in the gas can absorb photons — but only photons whose energy exactly matches the energy difference between two levels. The electron absorbs the photon and jumps to a higher energy level.

The photons of those specific wavelengths are removed from the beam of white light. When the transmitted light is spread by a prism or diffraction grating, the result is an absorption spectrum: a continuous rainbow of colours crossed by dark lines at exactly the same wavelengths as the bright lines in the element's emission spectrum.

The wavelengths of absorption lines and emission lines for the same element are identical. This is because both processes involve the same energy level differences — emission involves an electron dropping between two levels, absorption involves the same electron jumping between the same two levels in the opposite direction.

> **Key Point:** The absorption and emission spectra of any element are complementary: wherever there is a bright emission line, there is a dark absorption line at the same wavelength. This is why sodium street lamps (emission) and the Fraunhofer D-lines in the solar spectrum (absorption) appear at the same 589 nm wavelength.

48.9 Spectral Lines as Atomic Fingerprints

The uniqueness of spectral line patterns for each element is one of the most powerful analytical tools in science. No two elements have the same set of energy levels, so no two elements produce the same pattern of spectral lines. A spectrum is as unique to an element as a fingerprint is to a person.

Spectral analysis was developed by Gustav Kirchhoff and Robert Bunsen in the 1850s. They showed that each element produces a unique spectrum when heated, and used this to discover two new elements — caesium (identified by its blue spectral lines) and rubidium (red lines) — by observing spectral lines that matched no known element.

The identification of elements by their spectra is called spectroscopy or spectral analysis. It works regardless of how far away the source is, because light carries the spectral information across any distance. This makes it the primary tool for determining the composition of stars, galaxies, nebulae, and the intergalactic medium.

> **Historical Note:** In 1868, the element helium was discovered not on Earth but in the Sun. During a solar eclipse, Pierre Janssen and Norman Lockyer observed a bright yellow line in the solar spectrum that matched no known element. They named it helium (from Helios, the Greek sun god). Helium was not found on Earth until 1895, when William Ramsay detected the same spectral lines in gas released by uranium ore.

48.10 The Bohr Model of Hydrogen

In 1913, Niels Bohr proposed a model of the hydrogen atom that, for the first time, correctly predicted the wavelengths of its spectral lines. Bohr combined Rutherford's nuclear model with two new postulates that he could not justify classically but that produced the right answers.

Bohr's postulates: (1) Electrons orbit the nucleus only in specific allowed circular orbits, called stationary states, without radiating energy. In these orbits, the angular momentum of the electron is quantised: $L = nh/(2\pi) = n\hbar$, where $n = 1, 2, 3, \ldots$ and $\hbar = h/(2\pi)$. (2) Radiation is emitted or absorbed only when an electron transitions between two allowed orbits. The frequency of the emitted photon is given by $hf = E_{upper} - E_{lower}$.

From these postulates, Bohr derived the energy levels of hydrogen:

$$E_n = -13.6\ \text{eV} / n^2$$

Bohr energy levels for hydrogen. $n = 1, 2, 3, \ldots$ is the principal quantum number. The radius of the nth orbit is $r_n = n^2 a_0$ where $a_0 = 0.0529$ nm is the Bohr radius (radius of the ground-state orbit).

The series of transitions to different lower levels produce different named series of spectral lines. Transitions ending at $n = 1$ produce ultraviolet lines (Lyman series). Transitions ending at $n = 2$ produce visible lines (Balmer series, which includes H-alpha at 656 nm). Transitions ending at $n = 3$ produce infrared lines (Paschen series).

Series	Lower level	Spectral region	Named after
Balmer	$n = 2$	Visible (and near UV)	Johann Balmer
Paschen	$n = 3$	Near infrared	Friedrich Paschen
Brackett	$n = 4$	Mid infrared	Frederick Brackett
Pfund	$n = 5$	Far infrared	August Pfund

Table 48.3 *Named spectral series of hydrogen, each corresponding to transitions ending at a specific energy level.*

Example — Balmer series wavelength Find the wavelength of the line produced when an electron drops from $n = 4$ to $n = 2$ in hydrogen. $E_4 = -13.6/16 = -0.85$ eV. $E_2 = -13.6/4 = -3.40$ eV. $\Delta E = -0.85 - (-3.40) = 2.55$ eV $= 2.55 \times 1.602 \times 10^{-19} = 4.09 \times 10^{-19}$ J. $\lambda = hc/\Delta E = (6.626 \times 10^{-34} \times 3.00 \times 10^{8})/(4.09 \times 10^{-19}) = 486$ nm. This is the blue-green H-beta line.

48.11 Limits of the Bohr Model

The Bohr model was a triumph: it predicted hydrogen's spectral lines to within 0.02% accuracy. But it had serious limitations that became apparent immediately.

- It works only for hydrogen. For helium (two electrons), the Bohr model gives wrong predictions because it ignores electron–electron interactions. For multi-electron atoms, it fails entirely.
- It cannot explain the relative intensities of spectral lines — why some transitions produce bright lines and others produce faint ones.
- It cannot explain fine structure: with a more precise spectrometer, each hydrogen line turns out to be a pair of very closely spaced lines. The Bohr model predicts a single line.
- It cannot explain the Zeeman effect: spectral lines split when the atom is placed in a magnetic field. The Bohr model cannot account for this.
- It has no theoretical justification. Bohr simply assumed that angular momentum is quantised without explaining why. The true explanation requires quantum mechanics.

The complete explanation came with quantum mechanics (1925–1926), which replaced Bohr's orbits with probability distributions (electron clouds), introduced four quantum numbers to describe each electron state, and correctly predicted all atomic spectra, chemical bonding, and the periodic table. The Bohr model is a useful stepping stone but should not be taken as a literal picture of the atom.

48.12 Spectroscopy in Astronomy and Chemistry

Spectroscopy is the analysis of light using its spectral properties. It is the single most powerful diagnostic technique in science, applied across astronomy, chemistry, physics, medicine, and industry.

Astronomical spectroscopy

The composition of stars and galaxies is determined by analysing their spectra. The absorption lines in a star's spectrum (produced as light from the hot surface passes through cooler outer layers) reveal which elements are present. The Sun's spectrum contains over 25,000 absorption lines from more than 60 elements.

The Doppler effect shifts spectral lines to longer wavelengths (red shift) if the source is moving away from the observer, and to shorter wavelengths (blue shift) if approaching. By measuring the shift of known spectral lines, astronomers can determine the velocity of stars and galaxies. Hubble's discovery that distant galaxies show red shifts proportional to their distance was the key evidence for the expanding universe.

The width of spectral lines reveals the temperature of the gas (thermal broadening), the density of the gas (pressure broadening), and the rotation of stars. Even the magnetic fields of stars can be measured via the Zeeman splitting of spectral lines.

Chemical and laboratory spectroscopy

Flame tests (the basis of fireworks colours) use the characteristic emission colours of metal ions to identify elements: sodium → yellow, potassium → violet, copper → green, lithium → red. Modern analytical instruments (spectrophotometers, atomic absorption spectrometers) can identify and quantify elements at concentrations of parts per billion.

In organic chemistry, infrared spectroscopy identifies molecular bonds; nuclear magnetic resonance (NMR) spectroscopy identifies molecular structure; mass spectrometry identifies molecular masses. These tools underpin drug discovery, food safety testing, environmental monitoring, and forensic science.

> **Study Tip:** When an exam question gives a wavelength and asks which transition produced it, first convert to energy using $E = hc/\lambda$, then find which pair of hydrogen energy levels ($E_n = -13.6/n^2$ eV) has a difference equal to that energy. Round to two significant figures in the energy to account for calculation precision.

48.13 Common Mistakes in Atomic Spectra

- Confusing emission and absorption spectra. Emission: hot gas emits bright lines on a dark background. Absorption: white light passes through cool gas; dark lines appear at the same wavelengths in an otherwise continuous spectrum. Same wavelengths, different appearance.
- Using the wrong sign convention for energy levels. Hydrogen energy levels are negative (the electron is bound). The energy of an emitted photon is $E_{upper} - E_{lower}$, which is positive because the upper level has less negative energy. Always subtract the lower (more negative) from the upper (less negative).
- Forgetting to convert electron-volts to joules. The energy level formula gives eV; Planck's equation $E = hf$ uses joules. Multiply by 1.602×10^{-19} to convert eV to J, or divide by the same factor to go the other way.
- Applying the Bohr model to atoms other than hydrogen. The formula $E_n = -13.6/n^2$ eV applies only to hydrogen (one electron, one proton). For hydrogen-like ions (e.g. He^{+}, Li^{2+}) the formula becomes $E_n = -13.6\, Z^2/n^2$ eV, where Z is the atomic number. For multi-electron atoms, the formula does not apply at all.
- Confusing photon energy with photon intensity. A high-energy photon (short wavelength, high frequency) is not the same as a bright light. Brightness depends on the number of photons per second; energy per photon depends on frequency. A single gamma-ray photon has far more energy than a single red-light photon, but a dim gamma-ray source can be less intense than a bright red lamp.

Common Mistake: The most common exam error: calculating ΔE as $E_{lower} - E_{upper}$ and getting a negative photon energy. Photon energy is always positive. Calculate $\Delta E = E_{upper} - E_{lower}$ (which is positive for emission). If your ΔE comes out negative, you have subtracted in the wrong order.

Chapter Summary

Atomic structure and spectral lines reveal that energy inside atoms is quantised — restricted to discrete allowed values. The patterns of spectral lines are unique to each element and provide direct evidence for the existence of energy levels.

- Atoms have a tiny dense nucleus (protons and neutrons) surrounded by electrons. The nuclear model was established by Rutherford's gold-foil scattering experiment (1911).
- Electrons in atoms occupy discrete energy levels, not a continuous range. The ground state is the lowest energy level; excited states have higher energies.
- An electron emits a photon when it drops to a lower level; the photon energy = $E_{upper} - E_{lower}$. An electron absorbs a photon when it jumps to a higher level; the photon energy must exactly match the level difference.
- Emission spectra: bright lines on a dark background; produced by excited gas. Absorption spectra: dark lines on a continuous background; produced when white light passes through cool gas.
- The Bohr model correctly predicts hydrogen energy levels: $E_n = -13.6/n^2$ eV. Named spectral series (Lyman, Balmer, Paschen) correspond to transitions ending at n = 1, 2, 3 respectively.
- Spectral line patterns are unique atomic fingerprints used to identify elements in stars, laboratories, and on Earth.
- The Bohr model works only for hydrogen. Full quantum mechanics is required for multi-electron atoms.

Key Terms

atomic number (Z): Number of protons in the nucleus; equals the number of electrons in a neutral atom; defines the element.

nucleus: The dense, positively charged central region of the atom, containing protons and neutrons; about 10^{-15} m in diameter.

energy level: One of the discrete allowed energies an electron may have in an atom; indexed by principal quantum number n.

ground state: The lowest energy state of an atom ($n = 1$ for hydrogen); the most stable configuration.

excited state: An energy state above the ground state ($n > 1$); unstable; electron will eventually drop to a lower level, emitting a photon.

quantised: Restricted to specific discrete values; the energy of electrons in atoms is quantised — only certain values are allowed.

photon: A quantum (packet) of electromagnetic radiation with energy $E = hf = hc/\lambda$.

emission spectrum: The pattern of bright spectral lines produced when excited atoms emit photons as electrons drop to lower energy levels.

absorption spectrum: The pattern of dark lines in a continuous spectrum where photons have been absorbed by atoms, exciting electrons to higher levels.

Bohr model: Bohr's 1913 model of the hydrogen atom with quantised circular orbits; correctly predicts hydrogen energy levels as $E_n = -13.6/n^2$ eV.

Balmer series: Hydrogen spectral lines in the visible range produced by transitions from $n \geq 3$ down to $n = 2$.

spectroscopy: The study and analysis of spectra to determine composition, temperature, velocity, and other properties of matter.

ionisation energy: The energy required to remove an electron from an atom in its ground state; for hydrogen, 13.6 eV.

Bohr radius (a_0): The radius of the ground-state orbit in the Bohr model of hydrogen; $a_0 = 0.0529$ nm.

Concept Review Questions

Answer in complete sentences. No calculations required unless stated.

1. Explain why classical physics predicted that atoms should be unstable (electrons spiralling into the nucleus). What experimental evidence showed that atoms are, in fact, stable?
2. Describe Rutherford's gold-foil experiment: what was done, what was expected according to the Thomson model, and what was actually observed. What model did Rutherford propose to explain the results?
3. Explain the difference between the ground state and an excited state of an atom. What must happen for an electron to move from the ground state to an excited state? What happens when it returns to the ground state?
4. Why does a hot gas emit only specific wavelengths of light rather than a continuous spectrum? How does this relate to the concept of energy levels?
5. Explain how an absorption spectrum is produced. Why do the dark lines in a star's absorption spectrum appear at the same wavelengths as the bright lines in the emission spectrum of the same element?
6. What did Bohr have to assume (postulate) that classical physics could not justify? Why was the Bohr model considered a major advance despite being based on unjustified assumptions?
7. State three experimental observations that the Bohr model cannot explain. How does this show that the Bohr model, while useful, is not the complete picture?
8. Explain how astronomers determine the composition of a star from its spectrum. What additional information, beyond composition, can be obtained from spectral line analysis?

Atomic Model Questions

9. Draw and label a diagram of the nuclear model of the atom for a carbon atom ($Z = 6$, $A = 12$). Show protons, neutrons, and electrons. Why is almost all the mass in the nucleus?
10. Rutherford's nucleus is 10^{-15} m in diameter; the atom is 10^{-10} m. (a) What is the ratio of atomic diameter to nuclear diameter? (b) If the nucleus were scaled up to 1 mm, how far away would the nearest electron be? (c) What does this tell you about the density of the nucleus compared to the atom as a whole?

11. Explain why Thomson's plum pudding model predicted only small deflections in Rutherford's experiment, while Rutherford's nuclear model predicts large deflections for particles passing close to the nucleus.
12. The Bohr model fixes electrons in specific orbits, while the modern quantum model describes electrons as probability clouds. List two pieces of evidence or reasoning that favour the probability cloud model over fixed orbits.

Spectra Interpretation Exercises

13. A gas emits spectral lines at 410 nm, 434 nm, 486 nm, and 656 nm. (a) Which element does this most likely correspond to? (b) What series of transitions produces these lines? (c) Which line has the highest photon energy?
14. An astronomer observes the absorption spectrum of a distant star and notices that all hydrogen spectral lines are shifted to longer wavelengths compared to a laboratory hydrogen lamp. (a) What does this shift indicate about the star's motion? (b) If the H-alpha line (normally 656 nm) is observed at 670 nm, is the star moving toward or away from Earth?
15. Two elements are mixed in a gas discharge tube. The emission spectrum shows lines at 589 nm (yellow doublet), 656 nm (red), and 486 nm (blue-green). Identify the two elements present and explain your reasoning.
16. A student says: "An absorption spectrum has dark lines where the emission spectrum has bright lines — so they are opposite spectra." Is this statement correct? Clarify what "opposite" means and what is actually true about the relationship.

Energy Level Problems

17. For hydrogen: (a) Find the energy (in eV) of the photon emitted when an electron drops from n = 5 to n = 2. (b) Find the wavelength in nm. (c) Which named series does this belong to, and is it visible light?
18. A photon of wavelength 122 nm is absorbed by a hydrogen atom in its ground state. (a) Find the photon energy in eV. (b) To which energy level does the electron jump? (c) Is this an emission or absorption event?

19. Find the three longest wavelengths of light in the Lyman series of hydrogen (transitions from $n = 2, 3, 4$ down to $n = 1$). State whether each is visible, ultraviolet, or infrared.
20. A hydrogen atom is in the $n = 4$ excited state. List all possible single-step downward transitions the electron could make. For each, calculate the photon wavelength emitted.
21. The ionisation energy of hydrogen is 13.6 eV. (a) What is the minimum photon frequency needed to ionise a ground-state hydrogen atom? (b) What wavelength does this correspond to? (c) Is this visible, UV, or X-ray radiation?

Applied Spectroscopy Analysis

22. A flame test shows a bright yellow emission. (a) Which element is most likely responsible? (b) If you observe this line with a spectrometer at 589 nm, what is the photon energy in eV? (c) What does the presence of this line tell you about the energy levels of that element?
23. An astronomy student observes that the calcium K-line (normally at 393.4 nm) appears at 399.7 nm in the spectrum of a distant galaxy. (a) Is the galaxy moving toward or away from Earth? (b) Calculate the fractional wavelength shift $\Delta\lambda/\lambda$. (c) For low velocities, $v/c \approx \Delta\lambda/\lambda$. Estimate the recession velocity of the galaxy in km/s.
24. Explain why the Fraunhofer lines (dark absorption lines in the solar spectrum) tell us that the Sun's outer atmosphere is cooler than its surface. What would happen to the Fraunhofer lines if the outer atmosphere were hotter than the surface?
25. A forensic scientist uses atomic absorption spectroscopy to test for lead in a water sample. Explain the principle of this technique: what light source is used, what happens in the sample, and what is measured to determine lead concentration.

Chapter 48 Checkpoint

Before moving on to Chapter 49, confirm you can do each of the following.

I can...	See Section
Explain why classical physics fails for atoms and what evidence forced a new model	48.1
Describe Thomson's plum pudding model and explain why it was replaced	48.2
Describe Rutherford's gold-foil experiment and explain the nuclear model	48.3
Define atomic number, mass number, and explain the role of electrons	48.4
Define energy levels and ground/excited states; use $E_n = -13.6/n^2$ eV for hydrogen	48.5–48.6
Calculate photon energy and wavelength for a given atomic transition	48.7
Explain absorption spectra and how they differ in appearance from emission spectra	48.8
Use spectral lines to identify elements (spectral fingerprints)	48.9
State Bohr's postulates and apply his model to hydrogen transitions	48.10
State three limitations of the Bohr model	48.11
Describe applications of spectroscopy in astronomy and chemistry	48.12
Avoid common errors in energy level and spectra calculations	48.13

Continue to Chapter 49: Quantum Ideas →